

PARAS Grammar School – First Term Exam

(Ali Raza bad near Ali Town Station Lahore)

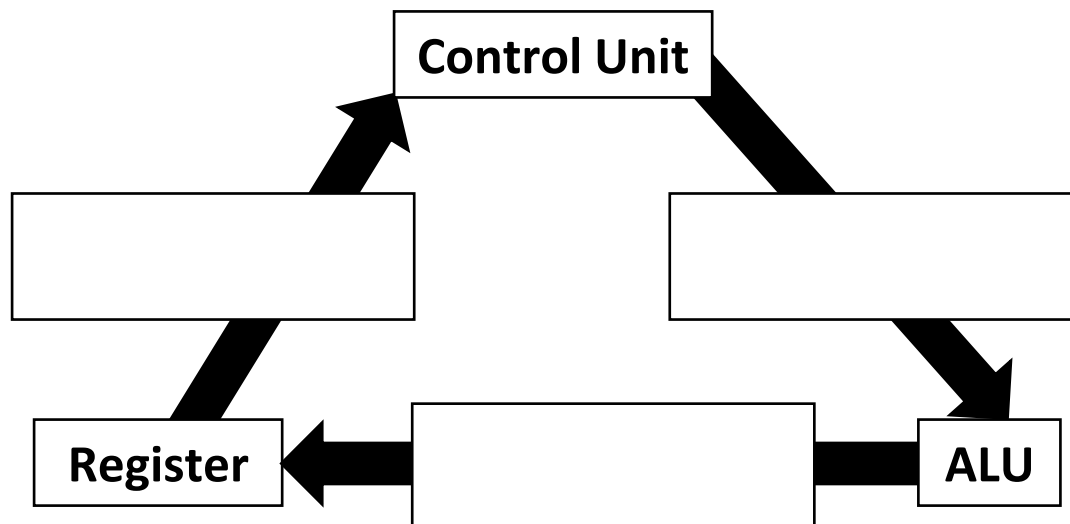
Course Title	Computer Science	Instructor/s	Mr. Aoun Haider		
Class	9 th	Syllabus	Chapter 1		
Time Allowed	2 hour	Date	19-05-2025	Total Marks	50
Student Name		Roll No			

Important Instructions/Guidelines:

1. Attempt all the questions
2. Solve all the questions on question paper except q no. 2, 3
3. Return the question paper along with your answer sheet

Q#01: Marks the correct option. [1x5 = 5 Marks] 1. Which bus among the following send the address of data? a) Address Bus b) Data Bus c) Control Bus d) Serial Bus 2. Von-Neumann architectures was developed in a) 1941 b) 1944 c) 1943 d) None of these 3. Hard disk (HD) is _____ memory. a) Transient b) Permanent c) Both (a) and (b) d) None of these 4. Process of designing, developing and maintaining software to perform certain tasks eradicating errors: a) Chemical Engineering Systems b) Physical Systems c) Social Systems d) Software Engineering Systems 5. Output Device a) Keyboard b) Mouse c) Printer d) Both (a) and (b)	Q#02: Attempt any five short questions: [2x5 = 10 Marks] 1. Define Systems? 2. Explain natural science and design science? 3. Define dynamic systems. 4. How study of algorithms involved in natural science of computer science? 5. What is Logic? 6. Write a brief note on Chemical Engineering Systems. 7. Define Artificial Systems. 8. Explain communication components in computer. Q#03: Long Question [1x10 = 10 Marks] 1. Write a detail note on working of CPU execution cycle.
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Q#04: Fill the empty boxes with name of the CPU execution state involved in the given diagram [5 Marks]



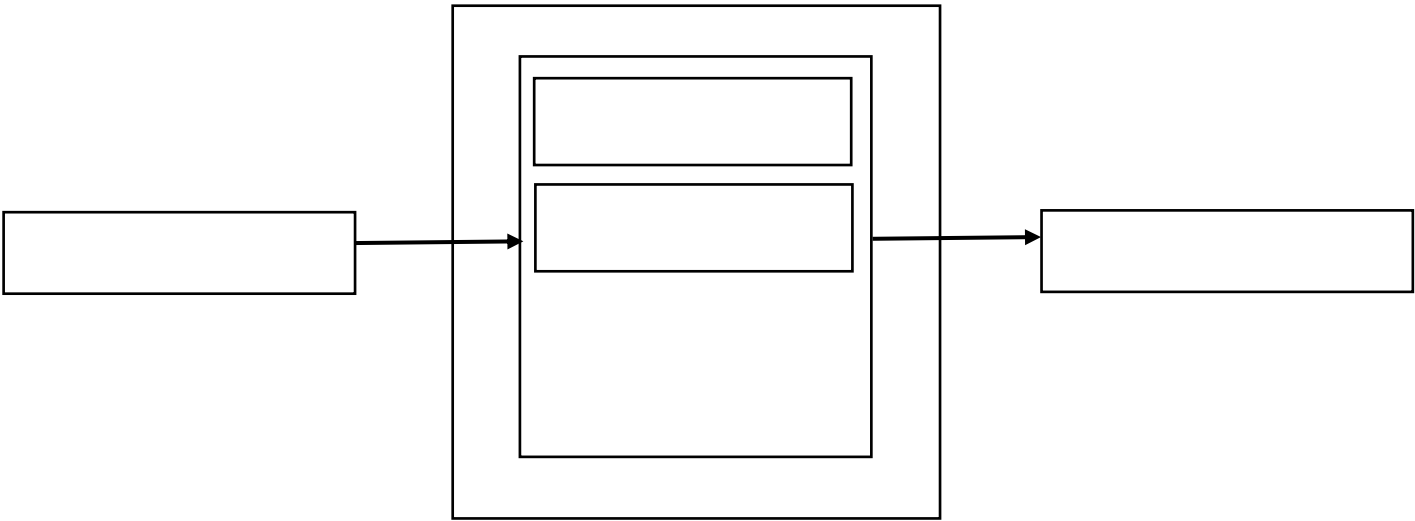
Q#05: Identify type of systems in the given scenario [1x10 = 10 Marks]
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A beehive operates with worker bees, drones, and a queen, each performing specific roles.	
A smart home adjusts lighting and temperature based on time of day and motion sensors.	
A student learns algebra by practicing problems and receiving feedback from a teacher.	

A car engine converts fuel into mechanical energy to move the vehicle.	
A virtual assistant answers voice commands and controls smart devices.	
A chemical reaction occurs when vinegar is mixed with baking soda, releasing gas.	
An online encyclopedia is constantly updated by a global community of users.	
A pendulum swings back and forth in a predictable motion under gravity.	
A social media platform connects people, enabling sharing of ideas and opinions.	
A bird navigates a migration route using instinct and environmental cues.	

Q#06: Fill the empty space boxes with the name of component in the given Von-Neumann architecture diagram

[5 Marks]



Q#07: Refer to Fig-1 and Identify CPU execution state in the given scenario [*Fetch, Decode, Execute, Store*]

[5 Marks]

The result of an addition is stored in memory address 200	
The CPU reads the next instruction from the program counter’s address.	
The decoded instruction indicates that two values must be compared.	
The ALU shifts a value two bits to the right.	
The CPU accesses RAM to get the data needed for an instruction.	